PROJECT DOCUMENTATION · PARTS 1–3

Supply Chain Analytics

Project Planning & Management
Literature Review
Requirements Gathering

TEAM

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TOOLS

Excel · SQL · Python · Power BI · Tableau

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Document Control

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GitHub repository	https://github.com/heshamrabei726-ux/-supply-chain-analysis
Planned deadline	4 December 2026

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Project at a Glance

Supply Chain Analytics is an end-to-end data analytics project. The team will clean, model and analyse the operational data of a supply-chain business – orders, customers, products, suppliers, warehouses, inventory and shipments – in order to measure performance, uncover problems and recommend improvements. Results are delivered as an interactive Power BI dashboard, cross-checked in Tableau, and supported by SQL and Python analysis.

22,000	7	2022–23	4	~7
orders	data tables	analysis period	team members	weeks to deliver

Headline figures computed from the raw files

Indicator	Value	Indicator	Value
Net revenue (excl. cancelled)	₹1,187.6 M	Late deliveries (delivered orders)	22.4 %
Active orders	20,477	Cancellation rate	6.9 %
Average order value	₹57,996	Shipping cost / net revenue	1.94 %
Gross margin	26.9 %	Stockout rate (inventory snapshots)	1.0 %

How to read this document

Part 1 explains how the project will be planned, staffed and controlled. Part 2 compares the project with a similar public project. Part 3 captures the functional and non-functional requirements the solution must satisfy.

PART 1

Project Planning & Management

Idea and objectives · scope · data and tools · team and RACI · timeline · risks · success KPIs

1.1 Project Idea and Overview

Supply-chain operations produce large volumes of data at every step: an order is placed, a supplier and warehouse fulfil it, a carrier ships it and the customer receives it. Without structured analysis it is hard to see where time and money are lost. This project turns the raw data into measurable KPIs, clear visuals and evidence-based recommendations.

Business problems the project addresses

- **Delivery delays** – 22.4% of delivered orders arrive after the promised date (average delay of late orders: 6.5 days).
- **Order cancellations** – 6.9% of orders (1,523) are cancelled before delivery.
- **Stock shortages** – 58 of 5,760 monthly inventory snapshots (1.0%) are flagged as stockouts.
- **Logistics cost** – shipping costs total ₹23.1 M (1.94% of net revenue); Air shipments cost about 2.5× more per shipment than Road.
- **Supplier variability** – supplier ratings range from 2.6 to 4.8 and lead times from 3 to 14 days.

1.2 Project Objectives

#	Objective	Measure of achievement	Target date
O1	Clean and validate all seven tables	100% of tables profiled; data-quality log completed	3 Nov
O2	Build a validated star-schema data model	All relationships tested; no ambiguous filter paths	6 Nov
O3	Define and calculate the project KPIs	KPI dictionary with 13 KPIs; totals reconciled	9 Nov
O4	Analyse delivery, supplier, inventory and cost performance with SQL and Python	Business questions answered with reproducible queries and notebooks	16 Nov
O5	Deliver an interactive Power BI dashboard and a Tableau cross-check	Dashboard covers all requirements (Part 3)	25 Nov
O6	Present insights and recommendations	At least 8 evidence-based insights; final documentation and presentation	4 Dec

1.3 Project Scope

In scope	Out of scope
Seven provided CSV tables, 2022–2023	Real-time or streaming data feeds
Data cleaning, validation and documentation	Building a transactional ordering website or mobile app (the data comes from such a system; our deliverable is the analytics solution)
Star-schema data model, DAX measures and KPI dictionary	Machine-learning forecasting (optional stretch goal only if time allows)
SQL and Python analysis of sales, delivery, suppliers, inventory and logistics cost	External data sources (market, weather, fuel prices)

In scope	Out of scope
Interactive Power BI dashboard and Tableau cross-check dashboard	Production deployment and scheduled data refresh
Documentation, GitHub repository and final presentation	

1.4 Dataset Overview

The project uses seven related tables. **orders** is the central fact table; **shipments** and **inventory** are additional fact tables; the remaining four tables describe customers, products, suppliers and warehouses.

Table	Records	Grain (one row =)	Role	Key columns
orders	22,000	One customer order line	Fact	order_id, customer_id, product_id, warehouse_id, supplier_id, sales_amount, order_status, promised / actual delivery dates
shipments	19,378	One shipment per shipped order	Fact	shipment_id, order_id, carrier, shipping_mode, shipping_cost, distance_km, delivery_status
inventory	5,760	Monthly stock snapshot per warehouse–product pair	Fact	inventory_id, warehouse_id, product_id, stock_quantity, reorder_level, stockout_flag
customers	2,000	One customer	Dimension	customer_id, segment, type, city, state, region
products	200	One product	Dimension	product_id, category, subcategory, unit_cost, selling_price, supplier_id
suppliers	10	One supplier	Dimension	supplier_id, region, rating, lead_time_days
warehouses	8	One warehouse	Dimension	warehouse_id, city, region, warehouse_capacity

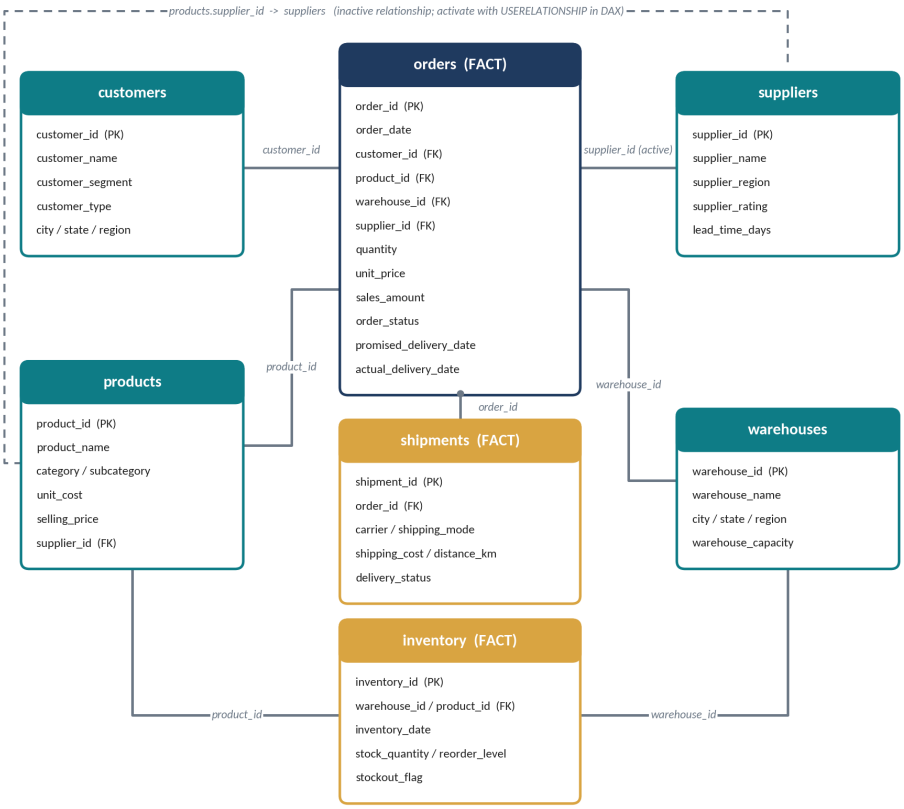


Figure 1 – Proposed star-schema data model (solid lines: active relationships; dashed line: inactive relationship).

Data observations from the initial profiling

- **No structural defects:** no duplicate rows, no orphan shipments, and *sales_amount* equals quantity × unit price in every order.
- **Missing values are expected, not errors:** 4,342 orders have no actual delivery date (Shipped 1,720 + Cancelled 1,523 + Processing 1,099), and 2,622 orders have no shipment record (exactly the Cancelled and Processing orders). Delivery KPIs must therefore use *Delivered* orders only.
- **Supplier ambiguity:** the supplier on the order differs from the supplier on the product in 19,882 of 22,000 orders (90.4%). The model must state which relationship is authoritative (see risk R1).
- **Stockout flag needs a documented rule:** 58 snapshots are flagged, while only 45 snapshots have zero stock.

1.5 Tools and Technologies

Tool	Purpose in the project	Primary users
SQL	Database creation, data loading, quality checks, KPI and business queries (joins, CTEs, window functions)	Abanoub; Malak, Hisham (cleaning)
Python	Cleaning scripts, exploratory analysis, feature engineering, statistical checks and charts (Pandas, Matplotlib / Seaborn, Jupyter)	Abanoub; Malak, Hisham (cleaning)
Excel	Initial profiling, pivot tables, quick KPI validation	Malak, Hisham
Power BI	Data model, DAX measures and the main interactive dashboard	Malak, Hisham, Mohamed Ramy
Tableau	Cross-check dashboard to validate figures and offer a second visual perspective	Mohamed Ramy, Abanoub
GitHub	Version control for SQL, Python, notebooks and documentation; one repository per team	All
Google Drive	Project tracking sheet (name and GitHub link), shared binary files (.pbix, .twbx, .xlsx)	All

Note: Tableau does not appear in the role list supplied by the team; its owners above are a proposal and can be changed.

1.6 Team Roles and Responsibilities

Member	Responsibilities	Main tools
Malak	Data cleaning (with Hisham) · DAX measures · Data modeling and relationships (with Hisham)	Power BI, Python, Excel
Hisham	Data cleaning (with Malak) · KPI definition and calculation · Data modeling and relationships (with Malak)	Excel, Power BI, Python
Mohamed Ramy	Insights and analysis · Power BI dashboard · Documentation and presentation	Power BI, Tableau
Abanoub	Querying, processing and analysing data with SQL and Python	SQL, Python

RACI matrix

R = Responsible (does the work) · A = Accountable (owns the result; one per task) · C = Consulted · I = Informed

Task	Malak	Hisham	Mohamed Ramy	Abanoub
Data cleaning	A/R	R	C	C
Data modeling and relationships	A/R	R	I	C
DAX measures	A/R	C	I	C
KPI definition and calculation	C	A/R	C	I
SQL and Python analysis	C	C	I	A/R
Insights and analysis	C	C	A/R	C
Power BI dashboard	C	C	A/R	C
Tableau cross-check dashboard	I	I	A/R	R
Testing and validation	R	A/R	R	R
Documentation and presentation	C	C	A/R	C

Proposed by the planner (not in the original role list): the Tableau task and the accountable owner for testing and validation.

1.7 Project Timeline

The plan assumes the work starts on **19 October 2026** and the final submission is on **4 December 2026** (about seven weeks). Each phase is linked to the owner and tool defined in the team roles. The plan keeps a one-week buffer between the finished dashboards (28 Nov) and the deadline for documentation, rehearsal and fixes.

Phase	Activities	Owner(s)	Tools	Start	End	Deliverable
A. Initiation	Kick-off, project name, GitHub repo and Drive sheet; Documentation Parts 1–3	All; Mohamed Ramy	GitHub, Drive	19 Oct	25 Oct	Parts 1–3 PDF; repo created
B. Data prep.	Database setup and loading; profiling; cleaning and data-quality log	Abanoub; Malak, Hisham	SQL, Python, Excel	20 Oct	3 Nov	Clean dataset and data-quality log
C. Modeling & KPIs	KPI dictionary; star-schema model; DAX measures	Hisham; Malak	Power BI, Excel	26 Oct	13 Nov	Data model, KPI dictionary, DAX library
D. Analysis	SQL and Python analysis; insight discovery and interpretation	Abanoub; Mohamed Ramy	SQL, Python	27 Oct	22 Nov	Query pack, notebook, insight list
E. Visualization	Power BI dashboard; Tableau cross-check dashboard	Mohamed Ramy; Abanoub	Power BI, Tableau	13 Nov	25 Nov	Dashboards
F. QA & delivery	Cross-tool validation; final documentation and presentation; rehearsal; submission	All; Mohamed Ramy	All	20 Nov	4 Dec	Final report, slides, repo

Gantt Chart

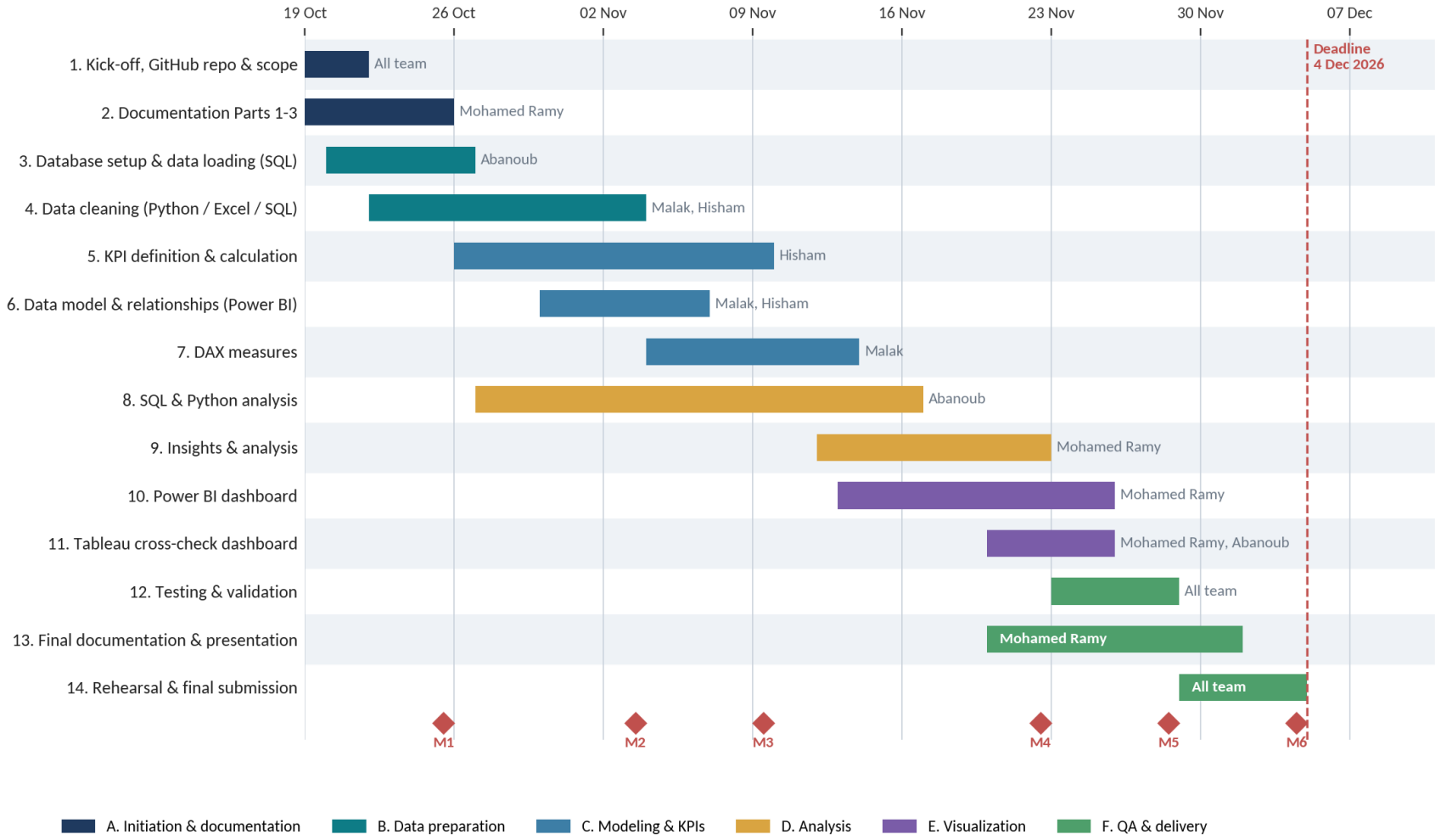


Figure 2 – Project Gantt chart with phases, owners and milestones (M1–M6). Deadline: 4 December 2026.

Milestones and acceptance criteria

ID	Date	Milestone	Acceptance criterion
M1	25 Oct	Documentation Parts 1–3 delivered	PDF reviewed by the team; repository and Drive entry completed
M2	3 Nov	Clean data ready	All seven tables cleaned; data-quality log signed off by Malak and Hisham
M3	9 Nov	Data model and KPIs locked	Relationships validated; KPI dictionary approved; no further definition changes
M4	22 Nov	Insights complete	At least 8 insights, each with evidence and a recommendation
M5	28 Nov	Dashboards validated	Power BI and Tableau totals reconcile with SQL and Python
M6	4 Dec	Final submission	Documentation, presentation and GitHub repository delivered

Team routine

- A 30-minute team meeting every week to review progress against the Gantt chart and to update the Drive tracking sheet.
- Code, SQL, notebooks and documents are committed to GitHub; binary files (.pbix, .twbx, .xlsx) are shared on Drive with a version suffix (v1, v2) and one editor at a time.
- Any change to a KPI definition goes through Hisham and is recorded in the KPI dictionary before it is used in any tool.

1.8 Risk Management

Each risk is rated by probability (P) and impact (I) on a 1–3 scale. Score = P × I: 1–3 Low, 4–6 Medium, 8–9 High.

ID	Risk	P	I	Score	Mitigation and response	Owner
R1	Supplier relationship is ambiguous: the order supplier differs from the product supplier in 90.4% of orders, which can distort supplier KPIs.	3	3	9	Treat orders.supplier_id as the fulfilling supplier (active relationship); keep products.supplier_id inactive and use USERELATIONSHIP when needed; document in the data dictionary; confirm with the supervisor.	Malak
R2	The reference project uses an identical dataset, so results may look similar and raise originality questions.	2	3	6	Cite the reference; differentiate through Tableau cross-check, DAX time intelligence, ABC inventory analysis and a supplier scorecard; validate every figure independently.	Mohamed Ramy
R3	Workload peak for one member (insights, dashboard, Tableau and documentation overlap 12 Nov – 1 Dec).	3	2	6	Shift Tableau co-ownership to Abanoub; Malak reviews DAX and visuals; Hisham supports documentation; start the final documentation early.	Mohamed Ramy
R4	KPI values differ between SQL, Python, DAX and Tableau because of different definitions.	2	3	6	Single KPI dictionary owned by Hisham; reconcile core totals across all tools at milestone M5.	Hisham
R5	Blank delivery dates and missing shipments are mistaken for errors and bias delivery KPIs.	2	2	4	Define the KPI population (Delivered orders only); document expected nulls in the data-quality log.	Hisham

ID	Risk	P	I	Score	Mitigation and response	Owner
R6	Tool access problems (Power BI Desktop, Tableau licence, database setup).	2	3	6	Verify installations by 21 Oct; use student or public licences; agree on a fallback machine.	Abanoub
R7	Binary files (.pbix, .xlsx) cannot be merged and may be overwritten.	2	2	4	One editor at a time; version-suffix naming; daily backup on Drive.	Malak
R8	Schedule slips and the 4 December deadline is missed.	2	3	6	One-week buffer before the deadline; weekly Gantt review; re-plan immediately if any milestone slips.	All
R9	Small differences between carriers, modes or warehouses are over-interpreted (late rates are all around 20–24%).	3	2	6	Report effect sizes and test significance before drawing conclusions; state limits of synthetic data.	Abanoub
R10	Insights are descriptive only and lack actionable recommendations.	2	2	4	Use the format metric – cause – recommendation; peer review at M4.	Mohamed Ramy

1.9 Success Measures and KPIs

Project success criteria

Criterion	Target	How it is measured
Schedule adherence	At least 5 of 6 milestones met within ±2 days	Gantt chart review at each weekly meeting
Data quality	100% of tables profiled; 0 unresolved critical issues	Data-quality log
Metric consistency	Core KPI totals identical across SQL, Python, DAX and Tableau	Reconciliation sheet at M5
Requirement coverage	100% of the requirements in Part 3 answered by the dashboard	Requirements list (Part 3)
Insight quality	At least 8 insights, each with evidence and a recommendation	Insight list reviewed at M4
Repository completeness	README, SQL scripts, notebooks, data dictionary and KPI dictionary committed	GitHub checklist
On-time delivery	Final documentation and presentation delivered by 4 Dec	Submission confirmation

Business KPIs and baselines

Baselines were computed directly from the raw files during planning and will be re-validated in Phase C.

#	KPI	Definition	Baseline
1	Net Revenue	Sum of sales_amount for non-cancelled orders	₹1,187,593,929
2	Active Orders	Count of orders with status other than Cancelled	20,477
3	Average Order Value	Net Revenue ÷ Active Orders	₹57,996
4	Gross Margin %	(Net Revenue − Σ quantity × unit_cost) ÷ Net Revenue	26.9 %
5	On-Time Delivery Rate	Delivered orders with actual date ≤ promised date ÷ Delivered orders	77.6 %
6	Late Delivery Rate	1 − On-Time Delivery Rate	22.4 %
7	Average Delivery Days	Mean of (actual delivery date − order date), Delivered orders	10.8 days
8	Average Delay of Late Orders	Mean of (actual − promised date) where late	6.5 days
9	Cancellation Rate	Cancelled orders ÷ all orders	6.9 %
10	Logistics Cost Ratio	Σ shipping_cost ÷ Net Revenue	1.94 %
11	Cost per Shipment	Σ shipping_cost ÷ number of shipments	₹1,191
12	Stockout Rate	Inventory snapshots with stockout_flag = 1 ÷ all snapshots	1.0 % (58 of 5,760)
13	Supplier Scorecard	Average rating and average lead time, with late-order share by supplier	3.5 / 5; 7.1 days

PART 2

Literature Review

Related project · comparison · strengths and gaps · lessons learned · reference

2.1 Related Project

The related project is **Supply Chain & Logistics Analytics**, a public GitHub repository by R. Jaiswal. It is an end-to-end analysis of supply-chain performance covering sales, inventory, suppliers, shipping costs and delivery performance. According to its README, the project analyses 22,000+ orders across 8 warehouses, 10 suppliers, 200 products and 5 regions for 2022–2023, answers 24 business questions, and uses SQL (MySQL), Python, Excel and Power BI. It was reviewed on 19 October 2026.

2.2 Comparison with Our Project

Both projects study the same business domain, use seven tables with the same names and the same record counts, and pursue the same questions: late deliveries, low stock, shipping cost and supplier performance.

Aspect	Reference project	Our project
Focus	Supply-chain and logistics performance	Same domain and business questions
Data	Seven tables, 2022–2023; 22,000 orders; 19,378 shipments; 5,760 inventory records; README states the data is synthetic	Identical structure, period and record counts (see check below)
Tools	SQL, Python, Excel, Power BI (repository description also mentions Tableau)	SQL, Python, Excel, Power BI and Tableau
Analysis	30+ SQL queries, Python EDA notebook, Excel analysis workbook	SQL and Python analysis plus a governed KPI dictionary and DAX measure library
Dashboard	Power BI, four pages: executive overview, logistics and delivery, supplier and warehouse, inventory and operations	Power BI dashboard driven by stakeholder requirements, plus Tableau cross-check
Team and documentation	Single-author portfolio project; README, methodology and business-insight reports	Four-member team with defined roles, RACI matrix, Gantt plan, risk register; formal documentation and final presentation

Dataset equivalence check

We recomputed the reference KPIs from our raw files; almost all match exactly, so both projects use the same dataset.

Metric	Reference README	Our computation	Match
Orders / shipments	22,000 / 19,378	22,000 / 19,378	Yes
Revenue / orders / AOV	₹1,187,593,929 / 20,477 / ₹57,996	₹1,187,593,929 / 20,477 / ₹57,996	Yes
Shipping cost / cost ratio	₹23,073,692 / 1.9 %	₹23,073,692 / 1.94 %	Yes
Late rate / days / delay	22.4 % / 10.8 / 6.5 days	22.4 % / 10.8 / 6.5 days	Yes
Cancellation / stockout	6.9 % / 1.0 %	6.9 % / 1.0 %	Yes
Gross margin	26.4 %	26.9 %	Differs

Gross margin differs by 0.5 points, probably because of a different cost definition; ours will be documented in the KPI dictionary (Phase C).

2.3 Strengths and Limitations of the Reference Project

Strengths	Limitations and gaps
• Complete end-to-end flow: data, cleaning, SQL, Python, Excel and Power BI • Clear business problem and five stated objectives • Star-schema design and a documented cleaning log • Prioritised recommendations (high, medium, low) • Reproducible: setup steps and repository structure are explained • Wide SQL coverage including CTEs and window functions	• Single author: no stakeholder analysis or formal requirements • Synthetic data limits real-world conclusions • Mostly descriptive analysis; little statistical testing of small differences • Forecasting, ABC analysis, supplier risk scoring and returns data are listed only as future work • No Tableau deliverable is documented • Gross margin definition is not fully explained

2.4 What Our Project Adds

- **Requirements-driven design:** every dashboard view is traced to a requirement (Part 3).
- **Cross-tool validation:** the same KPIs are reconciled in SQL, Python, Power BI (DAX) and Tableau, using one governed KPI dictionary and DAX library with time-intelligence measures.
- **Extended analysis:** ABC inventory classification, supplier scorecard, and significance checks on carrier and warehouse differences.
- **Explicit decisions and control:** documented supplier relationship and stockout rule, plus a Gantt plan, RACI matrix and risk register.

2.5 What We Learned from the Reference Project

- How to organise the data into fact and dimension tables and structure the repository (data, sql, python, excel, powerbi, reports).
- Which KPIs and questions matter: revenue, on-time delivery, delay, logistics cost ratio, stockout rate and supplier lead time; products with low stock, delayed deliveries, and shipping-cost differences between regions, modes and carriers.
- How to present results: a multi-page Power BI dashboard with an executive overview first, followed by detail pages.
- How to turn findings into prioritised, actionable recommendations.
- The team will use these ideas as a guide only and will carry out its own analysis, calculations and final dashboard.

2.6 Reference

Jaiswal, R. (n.d.). *Supply Chain & Logistics Analytics* [Source code repository]. GitHub. Retrieved 19 October 2026, from https://github.com/rajjais1232-coder/supply-chain-project

PART 3

Requirements Gathering

Functional requirements · non-functional requirements

3.1 Functional Requirements

The Supply Chain Analytics project aims to analyze supply chain data and provide meaningful insights through interactive dashboards and reports.

The main functional requirements are:

1. **Data Integration:** Import and combine supply chain datasets from the available data sources.
2. **Data Cleaning:** Identify and handle missing values, duplicate records, and data inconsistencies to improve data quality.
3. **Data Modeling:** Establish relationships between tables to support accurate and efficient analysis.
4. **KPI Calculation:** Calculate key performance indicators, including total sales, total profit, profit margin, order quantity, and shipping performance, based on the available dataset.
5. **Data Analysis:** Analyze sales, product performance, inventory-related information, and shipping trends where the required data is available.
6. **Interactive Dashboard:** Develop interactive Power BI dashboards that allow users to explore data using filters, charts, tables, and KPI cards.
7. **Reporting and Insights:** Present the main findings and provide insights that support data-driven decision-making.

3.2 Non-Functional Requirements

The project should meet the following non-functional requirements:

1. **Accuracy:** Calculations, KPIs, and visualizations should accurately represent the available data.
2. **Performance:** Dashboards and reports should respond efficiently to user interactions and filters.
3. **Usability:** The dashboard should have a clear, organized, and user-friendly design.
4. **Data Quality:** The dataset should be cleaned and validated before analysis to reduce errors and inconsistencies.
5. **Maintainability:** The data model, measures, and reports should be organized so that they can be updated and maintained when new data becomes available.
6. **Consistency:** All reports and visualizations should use consistent definitions for KPIs and business metrics.
7. **Security:** Access to project files and datasets should be managed appropriately, and sensitive information should be protected when applicable.